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Paleomagnetism of the Harp Lake Complex and associated rocks[1]

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The remanent magnetism of anorthositic rocks from the Harp Lake Complex consists of two antiparallel components, M and H, defined on the basis of their coercivities. M has remanent coercivities generally in the range 100–1000 Oe ($7.96 \times 10^3 - 7.96 \times 10^3$ A/m) but sometimes exceeding 2000 Oe (159×10^3 A/m), and is carried by magnetite and hematite, the former being the predominant contributor. H has remanent coercivities in excess of 2000 Oe (159×10^3 A/m), and is carried by hematite probably containing some ilmenite in solid-solution. The mean direction, irrespective of sign, based on 110 samples from 24 collecting sites is 270°, + 01 ° (α_{95} = 6°) with a pole 02° N, 154° W (A_{95} = 4°). This is considered to be the average direction of the geomagnetic field during initial cooling following intrusion of the complex at about 1450 Ma. At this time the complex was at great depth. The Harp dikes, which intrude the complex, have steeper inclinations, (263°, +44°, α_{95} = 7°, pole 19° N, 132° W). The country rocks in the thermal aureole have directions roughly similar to those of the complex, but deflected towards the directions in the dikes (272°, +29°, α_{95} = 10°, pole 14° N, 144° W). It is suggested that the country rocks were magnetized during final uplift of the area but prior to the intrusion of the Harp dikes. These and other paleomagnetic poles for the interval 1300–1500 Ma fall in the central Pacific indicating that Laurentia was in low latitudes. Previous reviewers have connected these poles by a simple polar loop, but these new observations indicate that the polar path may be more complicated, and a doubly looped path is suggested as a means of reconciling the results. There are about 20 accurately determined pole positions for the interval 1500–1200 Ma from North America, but only 4 from the rest of the world, and it is not yet possible to determine from such an unbalanced body of data the positions of other Precambrian shields relative to North America during this time interval.

Le magnétisme rémanent des roches anorthositiques du complexe de Harp Lake se résout en deux composantes antiparallèles, M et H, définies sur la base de leur coercibilité. M a une coercibilité rémanente généralement de l'ordre de 100 à 1000 Oe (7.96×10^3 à 79.6×10^3 A/m) mais excède quelquefois 2000 Oe (159×10^3 A/m), et elle est portée surtout par la magnétite, qui prédomine, et l'hématite. H a des coercibilités dépassant 2000 Oe (159×10^3 A/m) et est portée par de l'hématite qui contient probablement de l'ilménite en solution solide. La direction moyenne, faisant abstraction du signe et en se basant sur 110 échantillons provenant de 24 sites, est de 270°, +01° (α_{95} = 6°) avec un pôle à 02° N et 154° O (A_{95} = 4°). On considère cette valeur comme la direction moyenne du champ géomagnétique durant le refroidissement initial à la suite de l'intrusion du complexe il y a 1450 Ma. A ce moment, le complexe était à grande profondeur. Les dykes de Harp Lake, qui recoupent le complexe, ont des inclinaisons plus abruptes (263°, +44°, α_{95} = 7°, pôle à 19° N et 132° O). Les roches encaissantes à l'intérieur de l'auréole thermique ont des directions à peu près semblables à celles du complexe, mais dévient vers les directions dans les dykes (272°, +29°, α_{95} = 10°, pôle à 14° N et 144° O). On suggère ici que la magnétisation des roches encaissantes s'est produite durant le soulèvement final de la région mais avant l'intrusion

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des dykes. Ces données en plus d'autres pôles paléomagnétiques pour l'intervalle d'il y a 1300 à 1500 Ma indiquent le centre du Pacifique, ce qui laisse supposer que la Laurentie était alors à basses latitudes. Les articles de synthèse précédents ont relié ces pôles en une boucle polaire simple, mais ces nouvelles observations indiquent que la migration polaire a pu être plus complexe et une double boucle semblerait mieux expliquer les résultats. Il y a environ 20 positions polaires déterminées avec précision dans l'intervalle de 1500 à 1200 Ma à partir de l'Amérique du Nord, mais seulement quatre pour le reste du monde, et il n'est pas encore possible de déterminer à partir d'une telle disparité des résultats la position des autres boucliers précambriens par rapport à l'Amérique du Nord dans cet intervalle de temps.

[Traduit par le journal]

Introduction

Paleomagnetic studies of Precambrian rocks from North America carried out during the past 15 years, and especially during the past 5 years, have allowed reasonable estimates to be made of the path of polar wander during much of Proterozoic time. This path is the signature of Precambrian continental drift. Although there is plenty of scope for differences of opinion on details, it is now clear that the path consists of a number of loops which may be thought of as a series of curvilinear tracks separated by sharp bends or hairpins. Each loop has a duration of 250 Ma or more. The true situation is no doubt much more complex. These apparently simple tracks will probably be found eventually to be much more irregular than they appear at present and in this paper we present results which indicate that between about 1500 and 1250 Ma subordinate loops are present.

Geology of the Harp Lake Complex

The Harp Lake Complex is an Elsonian pluton lying for the most part within the Western Nain Subprovince of Stockwell (1964). It is one of several large complexes of anorthositic and related rocks lying north of the Grenville Front in central Labrador (Fig. 1). The other main complexes are the Michikamau Intrusion, the Mistastin pluton, and the Nain Anorthosite. Extensive paleomagnetic results are available from the Michikamau Intrusion and a reliable pole has been obtained (Murthy *et al.* 1968; Murthy *et al.* 1971; Emslie *et al.* 1976). A paleomagnetic survey of the Mistastin pluton has recently been described (Fahrig and Jones 1976). Preliminary results from the Nain Anorthosite have been reported by Murthy (1975) and by Dymek and Hargraves (1973).

The Harp Lake Complex covers about 10 000 km^2, and consists of anorthositic rocks including leucotroctolite, leucogabbro, anorthosite, and gabbro which occupy about 75% of the area, and adamellitic rocks which range from diorite through granite and make up the remaining 25%. The anorthositic rocks are commonly layered, and although layering attitudes with low dips are concordant over large areas, the layering does not define a simple overall coherent structure. The adamellites are massive and show intrusive relations toward the anorthositic rocks.

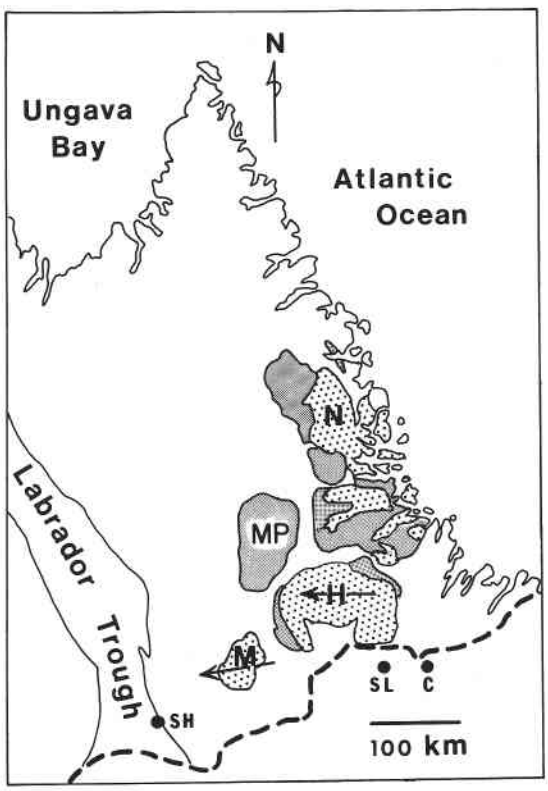


FIG. 1. Sketchmap showing the major plutons of Labrador: *M* Michikamau Intrusion, *H* Harp Lake Complex, *N* Nain Anorthosite, *MP* Mistastin pluton. The stippled bodies are anorthosites, the finer shaded areas are the associated adamellites. The arrows are the mean declination observed in the Michikamau and Harp Lake bodies. *SL*, *C* and *SH* are the Seal Lake Group, Croteau Group, and Shabagamo gabbro respectively. The Grenville Front is the dashed line.

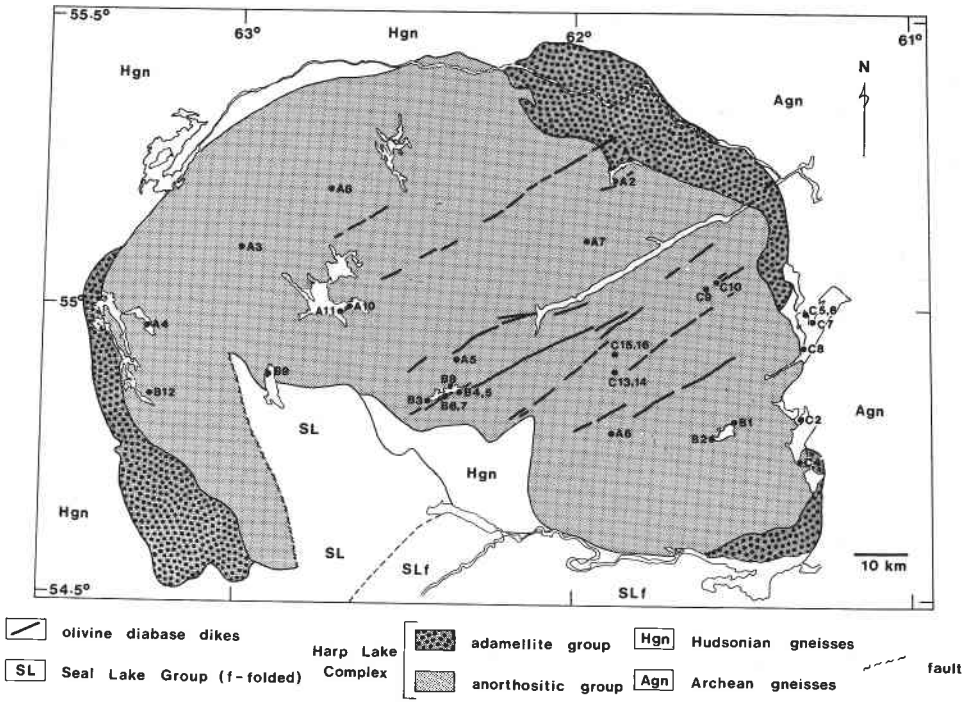


FIG. 2. Harp Lake Complex compiled from Emslie *et al.* (1972) and unpublished data. The sampling sites are labelled with the letters *A*, *B*, and *C* denoting that collections were made in 1970, 1971, and 1973 respectively.

No regional metamorphic overprint is present. The complex lies astride the boundary between a belt of medium-grade to high-grade metamorphic gneisses of Archean age to the east, and a younger belt of Hudsonian gneisses of similar metamorphic grade to the west (Fig. 2).

Igneous rocks of the anorthosite–adamellite suite in central Labrador have yielded K–Ar biotite and hornblende ages from 1510 to 1138 Ma. The few U–Pb zircon dates available suggest that the older part of this range (1500–1400 Ma) represents the main period of igneous emplacement and crystallization, and that the younger K–Ar dates are the result of differential cooling and uplift, and perhaps other, as yet unrecognized, thermal events.

A number of isotopic dates on rocks of the Harp Lake Complex itself, and its immediate envelope, have recently been determined. Hornblende from amphibolite in the contact aureole at the east margin gave a K–Ar age of 1449 ± 44 Ma (GSC, unpublished). Also, at the eastern contact, an igneous hornblende from marginal gabbro of the complex yielded a K–Ar age of 1482 ± 45 Ma (GSC, unpublished). A nearly concordant U–Pb zircon age of 1450 Ma (± a few Ma) was determined on a sample from the northeast adamellite mass by Krogh and Davis (1973). Biotite from the same mass gave a K–Ar age of 1193 ± 32 Ma (GSC, unpublished). The small granitic mass in the extreme southeast, which is included with adamellite group rocks in Fig. 2, has yielded a nearly concordant U–Pb zircon age of 1426 Ma (R. K. Wanless, personal communication, 1976) and a K–Ar hornblende age of 1351 ± 42 Ma (GSC, unpublished). Finally, a K–Ar whole rock age of 1260 Ma has been determined on anorthosite in the central part of the complex. These results indicate that the bulk of the complex completed crystallization at about 1450 Ma and that possibly some younger granites (southeast mass) were emplaced several tens of millions of years later, perhaps as a separate event. The younger K–Ar ages of 1193, 1351, and 1260 Ma may reflect later uplift, but they may also have been influenced by thermal disturbances associated with emplacement of the Harp dikes, and perhaps the Seal Lake igneous activity.

The Seal Lake Group, which has been studied

paleomagnetically by Roy and Fahrig (1973), occurs to the south of the complex and consists of red sediments, basic volcanic flows, and sills. These rocks are undisturbed in the northwest, but to the south and east (marked *f* in Fig. 2) they are strongly folded into a major synclinorium. The Seal Lake Group overlies the Harp Lake Complex and igneous rocks from it have yielded a Rb–Sr isochron of 1350 ± 92 Ma, $\lambda = 1.39 \times 10^{-11}$ years^{-1} (Baragar *in* Wanless and Loveridge, in preparation).

Magnetism of the Harp Lake Complex

The paleomagnetic sampling sites are marked on Fig. 2. Samples were obtained during the field seasons 1970, 1971, 1973. Usually 5 cores were obtained at each site, and from most cores two cylindrical specimens were cut. The magnetization was measured by Schonstedt SMI and DSM magnetometers. Alternating field (af) and thermal demagnetization were carried out using the apparatus described by Roy *et al.* (1971, 1973). All demagnetizations were carried out in ambient fields of less than 2 γ (nT) using the automatic field cancellation system described by Roy *et al.* (1969).

Paleomagnetic studies of the Michikamau Intrusion (Murthy *et al.* 1968; Emslie *et al.* 1976) have shown that single magnetizations with a westerly direction predominate. Subordinate antiparallel easterly magnetizations are, however, sometimes present. In contrast, in the Harp Lake Complex, easterly and westerly magnetizations occur with roughly equal frequency. Although rocks with single magnetizations do occur, most of the specimens studied contain two antiparallel magnetizations. These magnetizations are referred to as *M* and *H*, and are defined on the basis of their coercivities. The *M* magnetizations have moderate remanent coercive forces generally in the range 100–1000 Oe (7.96×10^3–79.6×10^3 A/m)[2], but sometimes exceeding 2000 Oe (159×10^3 A/m). For the most part they are carried by magnetite, with a subordinate contribution from hematite. The *H* magnetizations have much higher coercive forces, commonly exceeding 2000 Oe (159×10^3 A/m), and are due to hematite. For reasons given later, it was possible at each site to accurately separate only one of the magnetizations present. Five examples illustrating the range of behaviour are now given.

[2]1 Oe = 79.6 A/m in SI units.

Adamellite A1 (Fig. 3)

The adamellite is the youngest member of the complex and exhibits the simplest form of behaviour. Remanent coercities up to 1000 Oe (79.6×10^3 A/m) are present, and the thermal decay curve has two steps. The first step corresponds to blocking temperatures between 100 and 300°C, and is presumably caused by pyrrhotite which is abundant in polished sections. The second step corresponds to blocking temperatures between 500 and 600°C, and presumably is caused by magnetite, which is observed to occur as large grains (up to 500 μm) subdivided internally by cracks and veinlets. The directions of magnetization move away from the present geomagnetic field during the early stage of demagnetization, so that the low blocking temperature – low coercive force component is probably a viscous component produced by the present field. There is only a single stable magnetization directed toward the east with low inclination. The direction of this magnetization can be obtained by magnetic cleaning in fields of 1000 Oe (79.6×10^3 A/m), and thermal cleaning at 550°C. This is a typical *M*-type magnetization.

Anorthosite B5

Much more complex behaviour is exhibited by the anorthositic rocks. At site B5 (Fig. 4) the initial remanence (natural remanent magnetiza-

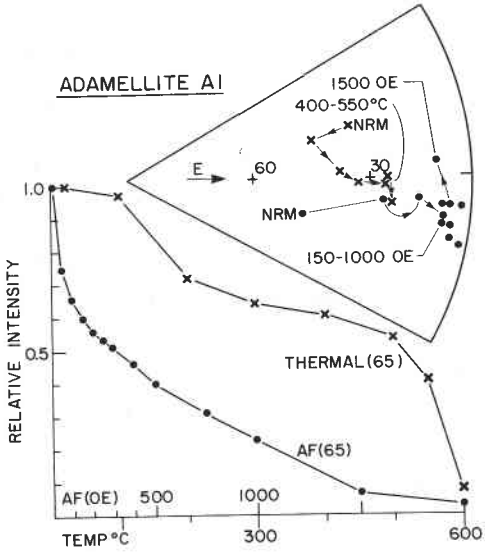


FIG. 3. The decay of intensity during af and thermal demagnetization site A1. The initial intensities in microgauss are shown in brackets. The corresponding directional variations are shown on an equal-area projection in the inset as crosses (thermal) and dots (af).

tion or NRM) is very strong (28 816 microgauss) and is directed toward the east. During af demagnetization the direction migrates toward the west, but does not reach a satisfactory end-point in fields of up to almost 2900 Oe (230.8×10^3 A/m), the maximum alternating field available (Roy *et al.* 1973). The intensity falls rapidly and levels off to about 1.5% of the initial value. The remanence consists of two stable magnetizations. There is a very strong easterly magnetization, comprising over 98% of the magnetization, which has remanent coercivities ranging up to 2900 Oe (230.8×10^3 A/m) but which are *predominantly* less than 300 Oe (23.9×10^3 A/m) (an M magnetization). This M magnetization is very probably carried mostly by magnetite, which is present as numerous minute particles and rods in the feldspars, as large (500 μm) grains with irregular veins and patches, and as inclusions in large grains of ilmenite. There is also a westerly magnetization with remanent coercivities an order of magnitude greater, and intensity 50 times less (an H magnetization). Between 5 and 10% of the rods in the feldspars are hematite, and these are very probably the carriers of the H magnetization. Either the NRM itself, or the remanence after af demagnetization in 100 or 200 Oe (7.96×10^3 or 15.92×10^3 A/m), provides a good estimate of the direction of the M magnetization. An accurate estimate of the direction of H cannot be made because cleaning is incomplete in the maximum demagnetizing fields available. Although most of the M magnetization is carried by magnetite, a small but significant proportion must have remanent coercive forces up to and exceeding 2900 Oe (230.8×10^3 A/m), otherwise it would have been possible to retrieve H by magnetic cleaning. Hence a very small part of M probably is carried by hematite. The remanent coercive force spectra of M and H overlap in the 300–2900 Oe (23.9×10^3–230.8×10^3 A/m) range.

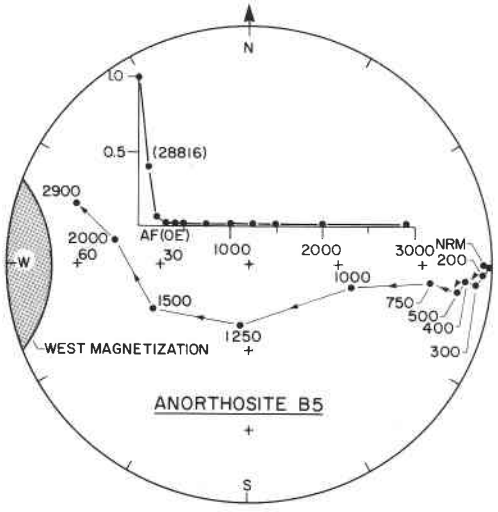


FIG. 4. Site B5. The directions (all with downward indications) are plotted on an equal-area projection and labelled with the appropriate strengths (peak) in oersteds. The corresponding intensity changes are shown in the inset. The very high initial intensity in microgauss is given in brackets. The zone within which the westerly magnetizations generally fall is shown.

Leucotroctolite A5 (Fig. 5)

The behaviour in af of specimens from A5 has some similarities to that already described for site B5. There is an excellent early end-point between 150 and 500 Oe, which characterizes the M magnetization, but the migration toward the west is delayed until higher fields. The directional changes are again incomplete at 2900 Oe (230.8×10^3 A/m). Once again M makes up over 98% of the magnetization, and although moderate remanent coercivities predominate and magnetite is presumably therefore the main carrier, hematite probably also contributes to the M magnetization, otherwise cleaning of the H magnetization would have been complete in fields of less than 2000 Oe (159.2×10^3 A/m), which it evidently is not. Thermal demagnetization of a second specimen yielded no change in direction up to 650°C (Fig. 5) and reveals only the M magnetization. Most of the magnetization has blocking temperatures between 550 and 600°C, but a small and significant proportion is retained at 650°C, indicating again that although magnetite is by far the major carrier of M, hematite also contributes to it. The opaque minerals are similar to those described for site B5. There are numerous and sometimes very long (500–1000 μm) oxide rods in the feldspars. Magnetite predominates but hematite also is present.

Leucogabbro A4 (Fig. 6)

The directional changes during af demagnetization at this site are the mirror image of those just described. The magnetization is now predominantly westerly. There is little change in direction in alternating fields up to 750 Oe (59.9×10^3 A/m), but in higher fields there is an easterly trend which is incomplete at 2900 Oe (230.8×10^3 A/m). The M magnetization, which

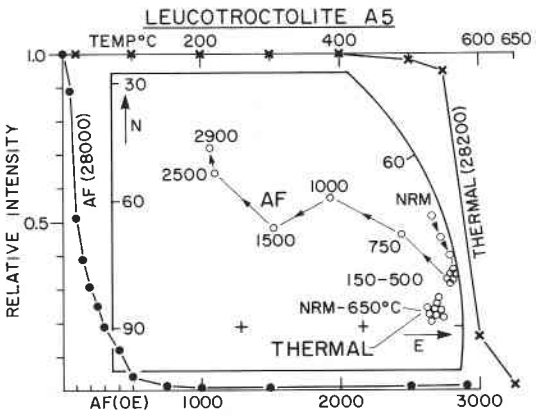


FIG. 5. Site A5. The decay of intensity during af (lower scale) and thermal (upper scale) demagnetization is shown, together with the corresponding direction variations in the inset. The directions all have upward inclination. The initial intensities are given in brackets in microgauss.

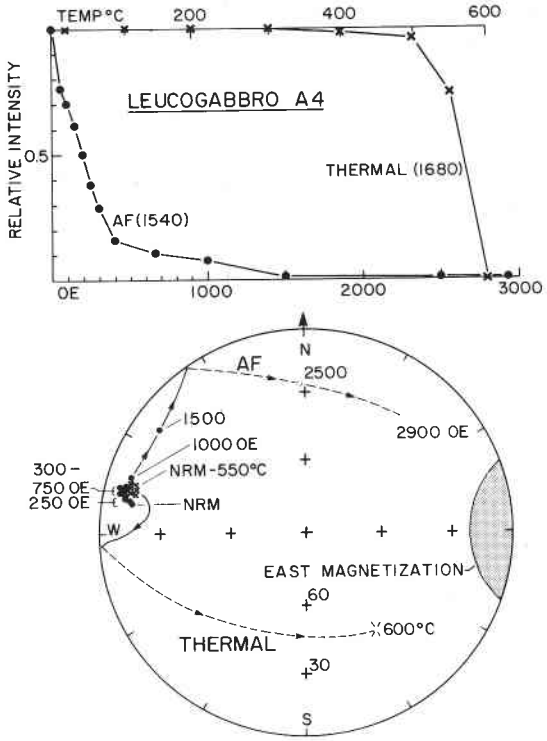


FIG. 6. Site A4. Above, the decay of intensity during af (lower scale) and thermal (upper scale) demagnetization is shown. The initial intensities are given in brackets. Below, the corresponding directional changes are shown. The zone within which the easterly magnetizations fall is shown.

now is directed toward the west, constitutes 99% of the intensity, has remanent coercive forces generally less than 1000 Oe (79.6×10^3 A/m), and is presumably carried by magnetite. The easterly *H* magnetization has a magnitude one hundred times less, remanent coercive forces exceeding 2900 Oe (230.8×10^3 A/m), and is presumably caused by hematite. Under thermal demagnetization there is negligible directional change up to 550°C, but at 600°C there is an easterly trend. Blocking temperatures are mainly between 550 and 600°C, indicating that magnetite is the carrier of most of the remanence. The opaque minerals are similar to those at site B5 except that the anisotropic hematite rods now make up about 20% of the rods in the feldspars.

Anorthosite A8 (Fig. 7)

At site A8, one specimen shows a complete 180° migration during af demagnetization with an excellent end-point in the range 1500–2900 Oe (119.4×10^3–230.8×10^3 A/m); the partial reversal documented in Figs. 4, 5, and 6 is here completed. The moderate coercive force easterly *M* magnetization comprises 92% of the remanence, and its progressive removal and replacement by the high coercive force *H* remanence is beautifully described by the decay curve with its minimum between 200 and 300 Oe (15.9×10^3 and 23.9×10^3 A/m) corresponding to the intermediate directions. During thermal demagnetization there are no directional changes up to 500°C, but between 500 and 650°C substantial fluctuations occur although the directions remain easterly. The blocking temperatures are predominantly between 500 and 600°C suggesting that magnetite is the dominant carrier of remanence, but 1.5% of the intensity is retained between 600 and 650°C indicating that hematite also contributes. The opaque minerals are as before, the hematite rods comprising about 10% of the rods in the feldspars.

Leucogabbro A10

In the examples described, *M* has been approximately two orders of magnitude greater than *H*. This is the most common situation. In some samples, however, notably from site A10 (Fig. 8), *H* constituted a larger proportion of the remanence, so that the NRM, instead of having an easterly or westerly direction, falls in an intermediate position. The more even balance between the *M* and *H* magnetizations is reflected in the opaque oxides. Hematite is now the most abundant oxide, occurring as grains (50 μm)

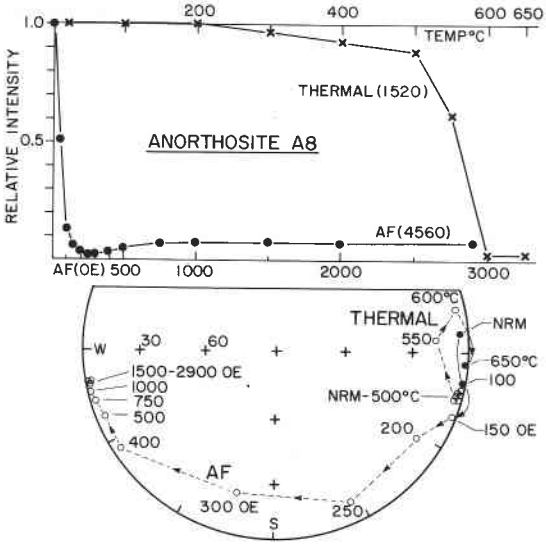


FIG. 7. Site A8. Above, the decay of intensity during af (lower scale) and thermal (upper scale) demagnetization is shown. The initial intensities are given in brackets. Below, the corresponding directional changes are shown.

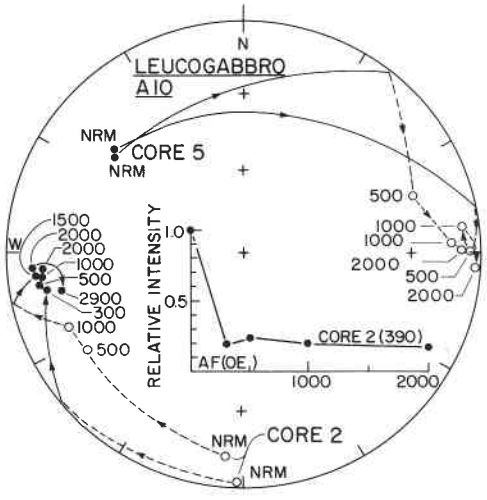


FIG. 8. Site A10. af demagnetization of two specimens from each of two cores. The corresponding intensity changes for one specimen of core 2 are shown in the inset. Circles (dots) are upward (downward) directions.

intergrown with ilmenite, and as rods in feldspars. The NRM of two specimens from core 5 has northwesterly directions and after demagnetization in 1000 Oe (79.6×10^3 A/m) they become easterly (Fig. 8). *H* has been successfully isolated. In core 2 the NRM directions in two specimens are southerly, and after demagnetization they become westerly with excellent end-points between 500 and 2900 Oe (39.8×10^3 and 230.8×10^3 A/m). Specimens from the same core are in excellent agreement. Both easterly and westerly directed *H* magnetizations, and their corresponding westerly and easterly *M* magnetizations, are present at the same site. The *H* magnetization at locality A10 can be isolated experimentally, and at first sight the direction variations shown in Fig. 8 might be taken to indicate that the *M* magnetizations could also be determined from the vector differences between successive demagnetization steps. In fact, however, successive vector differences yield no coherent directions, so that during demagnetization some of the higher coercive force magnetization is being removed; that is, the remanent coercive force spectra of the *H* and *M* magnetizations again overlap, so that vector analysis cannot be used to isolate the two magnetizations present. Only *H* can be retrieved. The reason for this is that hematite is present in both the *M* and *H* magnetizations.

Summary

The remanent magnetizations of the anorthositic rocks almost invariably consist of two antiparallel magnetizations, *M* and *H*. *M* is carried predominantly by magnetite with moderate remanent coercive force generally in the range 100–1000 Oe (7.96×10^3–79.6×10^3 A/m), but with a small contribution from hematite with coercive force commonly exceeding 2000 Oe (159.2×10^3 A/m). The *H* magnetization is carried by hematite. In different specimens *M* and *H* may be either easterly or westerly. In the same specimen *M* and *H* appear always to be antiparallel. The identification of *M* and *H* can only be made satisfactorily by af methods. It cannot be achieved by thermal demagnetization presumably because the hematite contribution to *M*, although small compared to that of magnetite, is generally larger than the hematite contribution to *H*. Only in a few instances (Fig. 7) can *M* and *H* be fully separated experimentally. Either *M* or *H* can be obtained, but not both. These two antiparallel magnetizations may have been caused by reversals of the field occurring during slow cooling of these coarse-grained plutonic rocks, or alternatively *H* magnetizations (in whole or in part) may have been induced by an internal field existing in earlier formed magnetite; the available information is insufficient to decide between these alternatives.

Analysis of Directions

At three sites in the complex no coherent directions could be found by either thermal or af methods (A9, B10, and C12), but elsewhere an accurate estimate of either M or H can be obtained. Usually M can be obtained accurately by cleaning in alternating fields up to 500 Oe (39.8×10^3 A/m) because M is very much larger than H (Figs. 4 and 5) and the effect of H can be neglected. When M is less dominant, H can usually be obtained directly by cleaning in high alternating fields (Fig. 8). The directions obtained in this way from all cores are plotted in the upper part of Fig. 9. The averages for each site are listed in Table 1, where the source (M or H) is noted. When antiparallel directions have been isolated at the same locality (Fig. 8), they have been averaged irrespective of sign. The directions of M and H magnetizations averaged separately (irrespective of sign) over the complex are not significantly different.

Harp Dikes

The Harp dikes are a swarm of vertical olivine diabase dikes that strike west-southwest across the Harp Lake Complex (Fig. 2). They are commonly 30–60 m wide, with chilled margins, and are unmetamorphosed. Chemically the rocks are transitional between alkalic and tholeiitic basalt. In this respect they are similar to the Seal Lake lavas and are believed to be more or less contemporaneous (Meyers 1974). There is some field and geochronological evidence for this opinion. Large slabs and blocks of red arkose and pebble conglomerate occur over a wide area in the western part of the Harp Lake Complex (near sites A10 and A11, Fig. 2). The rocks are friable and could not have withstood long-distance transport. A thin section of one of the blocks contains subangular to rounded labradorite crystal fragments, lithic pebbles of anorthosite, and two small rounded pebbles of fine-grained, oxidized but otherwise fresh diabase petrographically similar to the Harp dikes. If these red sediments are correlative with the Seal Lake Group red beds cropping out about 15 km to the south, and if the diabase pebbles are indeed from the Harp dikes, then the Harp dikes are either older than, or nearly contemporaneous with, Seal Lake Group sedimentation.

Two whole-rock K–Ar dates from the Harp dikes yielded 1316 ± 94 Ma and 1080 ± 82 Ma (GSC, unpublished). Whole-rock systems are notorious for losing argon and the older age is preferred; this is consistent with the age of Seal Lake igneous activity (1350 Ma, see above). The best current estimate of the age of Harp dike emplacement is therefore *about* 1350 Ma.

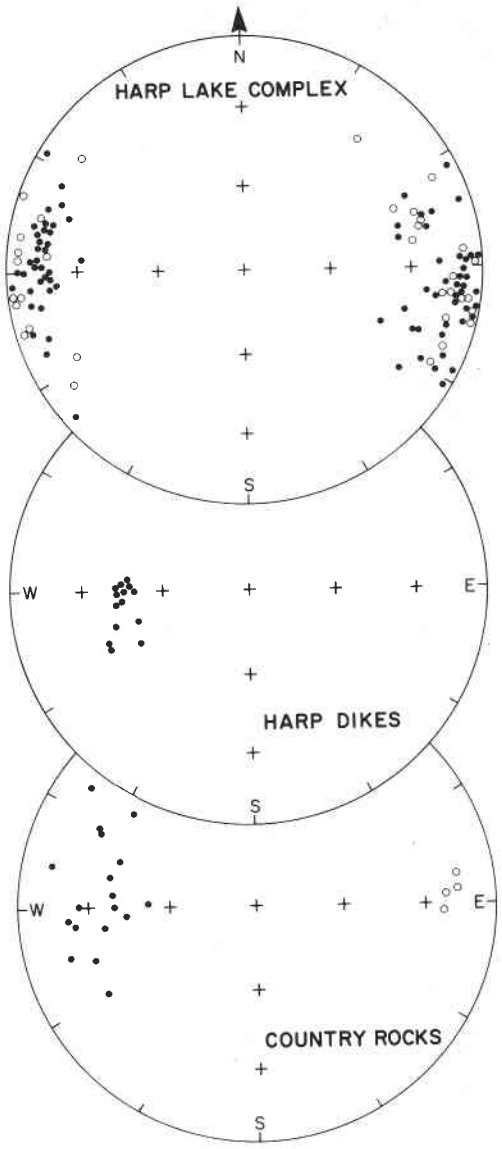


FIG. 9. Directions in all samples from the complex, the dikes, and the country rocks. Each direction is the mean of two specimens from one core sample. The primitive is the present horizontal plane. Open circles (solid dots) denote upward (downwards) directions.

The magnetization of the Harp dikes is comparatively simple. Cleaning in fields of a few hundred oersteds produces well-grouped directions (Fig. 9). The blocking temperatures are

TABLE 1. Results from collecting sites

	UTM Grid, Zone ZOU	N	J_N	$D°, I°$	R	$\delta°$	MAG	Treatment
Harp Lake Complex								
A1 Adamellite	E574000, N6124400	5	60	103, +08	4.88	13	M*	1000 Oe, 550°C
A2 Leucotroctolite	E570700, N6120200	5	458	097, +24	4.91	11	M* H	1000 Oe, 500°C
A3 Anorthosite	E501100, N6106400	5	2293	273, +11	4.92	10	M H*	2000 Oe, 1000 Oe, 650°C
A4 Leucogabbro	E483100, N6090600	5	1586	281, +13	4.98	6	M* H	500 Oe, 550°C
A5 Leucotroctolite	E544000, N6085500	5	13780	099, +05	4.88	13	M* H	500 Oe, 650°C
A6 Anorthosite	E571700, N6072400	3	3076	096, +16	2.85	18	M*	1000 Oe, 500 Oe, 550°C
A7 Leucogabbro	E566400, N610900	5	27922	072, +32	4.96	7	M* H	300 Oe, 650°C
A8 Anorthosite	E517400, N611760	5	1792	101, +00	4.59	23	M* H	600 Oe, 200 Oe, 100 Oe
A10 Leucogabbro	E521000, N6095600	5	1316	261, +01	4.66	21	M H*	2900 Oe, 1000 Oe, 500 Oe
A11 Anorthosite	E518400, N6094800	5	617	086, +08	4.83	15	M H*	2900 Oe, 2000 Oe, 1000 Oe
B1 Anorthosite	E594700, N6074300	5	2409	273, +11	4.95	8	M*	NRM
B2 Anorthosite	E590800, N6071500	5	7096	272, +13	4.73	19	M* H	300 Oe, 550°C
B3 Leucotroctolite	E536800, N6077400	4	12736	108, +08	3.90	13	M* H	200 Oe
B5 Anorthosite	E542400, N6079400	3	26177	097, +07	2.99	5	M* H	200 Oe
B8 Anorthosite	E540000, N6078800	5	9529	097, −07	4.85	14	M* H	300 Oe
B9 Anorthosite	E506100, N6082400	3	111	280, +20	2.97	8	H*	2000 Oe
B12 Leucotroctolite	E483800, N6077900	5	2166	276, +16	4.77	17	M* H	750 Oe, 550°C
C4 Marginal gabbro	E607400, N6067200	5	774	267, +14	4.94	9	M*	500 Oe
C9 Leucogabbro	E589200, N6100300	7	1428	271, −02	6.79	14	M* H	1000 Oe, 500 Oe, 300 Oe
C10 Leucogabbro	E591100, N6101600	5	1645	268, −13	4.63	22	M* H	500 Oe
C13 Anorthosite	E572000, N6083900	2	188	080, −06	1.88	20	M H*	2900 Oe, 2500 Oe
C14 Anorthosite	E571800, N6084700	3	368	101, −05	2.90	15	M* H	500 Oe, 300 Oe
C15 Leucogabbro	E571900, N6087200	5	3121	099, −05	4.70	19	M* H	300 Oe
C16 Anorthosite	E571700, N6088100	5	3457	071, −19	4.67	20	M*	500 Oe
Harp Dikes								
B4 Olivine diabase	E542400, N6079400	4	2983	263, +44	3.95	9		300 Oe
B6 Olivine diabase	E540700, N6078400	5	2604	256, +45	4.94	9		300 Oe
B7 Olivine diabase	E540800, N6078500	5	2842	269, +44	4.99	3		300 Oe
Country Rocks								
C2 Amphibolite (2.8 km)	E607800, N6076400	2	150	272, +33	1.99	7		500 Oe
C5 Amphibolite (2.5 km)	E607800, N6096200	4	18	087, −19	3.99	5		500 Oe, 300 Oe
C6 Amphibolite (2.6 km)	E607900, N6095800	5	420	274, +31	4.78	17		500 Oe, 300 Oe
C7 Amphibolite (4.0 km)	E609600, N6094100	4	1	285, +21	3.58	27		300 Oe
C7 Metadiabase (4.0 km)	E609600, N6094100	5	1398	279, +41	4.85	14		500 Oe, 300 Oe
C8 Amphibolite (0.4 km)	E607700, N6089200	2	343	258, +27	1.99	6		500 Oe, 300 Oe

NOTE: N is the number of separately oriented cores, 2 specimens cut and studied for each core. J_N is the intensity of NRM in microgauss. D^0, I^0 is the mean direction, R the resultant and δ^0 the standard direction. In the column headed MAG the presence of M and (or) H magnetization is shown; the dominant magnetization is indicated by an asterisk and its direction is given. The cleaning procedure used is indicated in the last column. The distances of country rock from contacts are indicated.

TABLE 2. Summary of results

		B	N	$D_m°, I_m°$	k	$\delta°$	$\alpha_{95}°$	Pole	dm, dp
Harp Lake Complex	normal directions	10	50	272.1, +8.5	49	11.0	7.0	4.7 N, 151.6 W	7.0, 3.6
	reversed directions	14	60	093.7, +4.8	22	17.0	8.7	0.2 S, 156.7 W	8.7, 4.4
	all directions	24	110	273.0, +0.8	25	16.0	6.1	2.1 N, 154.6 W	6.1, 3.0
	all poles	24	110	—	46	11.7	4.4	*1.6 N, 153.7 W*	
Country Rock	directions	6	22	272.4, +28.9	49	10.6	9.7	13.9 N, 144.1 W	10.7, 5.9
	poles	6	22	—	63	10.1	8.5	*14.1 N, 144.2 W*	
Harp Dikes	directions	3	14	263.0, +44.4	300	3.8	7.1	17.1 N, 131.0 W	9.0, 5.6
	poles	3	14	—	414	3.3	*6.1*	*18.6 N, 131.9 W*	

NOTE: B is the number of collecting sites and N the number of separately oriented core samples, $D_m°, I_m°$ is the mean direction and k (precision) $\delta°$ (standard deviation) and $\alpha_{95}°$ (circle of confidence) the statistics for the directions and poles. dm, dp are the semi-axes of the error oval around the pole. The italicized values are regarded as the best overall summaries.

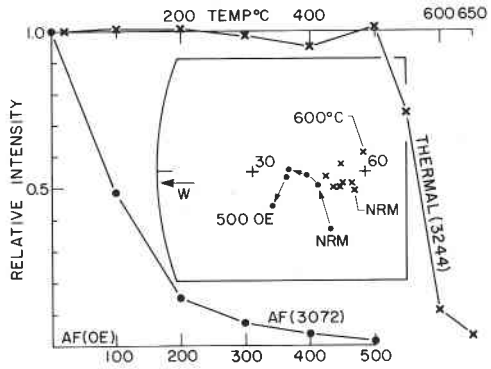


FIG. 10. The decay of intensity during af and thermal demagnetization in the Harp dike B7. The corresponding directions are also shown. At 650°C the magnetization direction scatters. The initial (NRM) intensities in microgauss are given in brackets.

high, predominantly between 500 and 600°C (Fig. 10). The directions are listed in Table 1.

Magnetism of Country Rocks

Samples were obtained from a variety of rocks, ranging from granite gneisses to amphibolite and metadiabase, between 0.4 and 4.0 km from the contact (Fig. 2). The granitic gneisses (sites C1 and C3) gave incoherent results as Park (1973) has found them to do elsewhere in the Canadian Shield. Samples from lenses of basic material, however, often gave coherent results with remanent coercivities generally less than 500 Oe (39.8×10^3 A/m), but occasionally ranging up to 1000 Oe (79.6×10^3 A/m). Magnetic cleaning in several hundred oersteds yielded directions that were predominantly westerly with inclinations about 20° steeper than those of the complex. These directions were usually retained during thermal demagnetization up to 550°C, and there was no migration toward the shallower inclinations of the complex. One site (C5) had easterly directions, which, during thermal demagnetization, remained easterly up to 350°C, but at 400°C changed dramatically to an antiparallel direction (260°, +29°), and remained there up to 550°C. This reversal may have been caused by reversal of the field during slow cooling of the rocks of the thermal aureole. The results are plotted in Fig. 9 and listed in Table 1.

Discussion

The overall mean directions and corresponding poles are given in Table 2. Unit weight is given

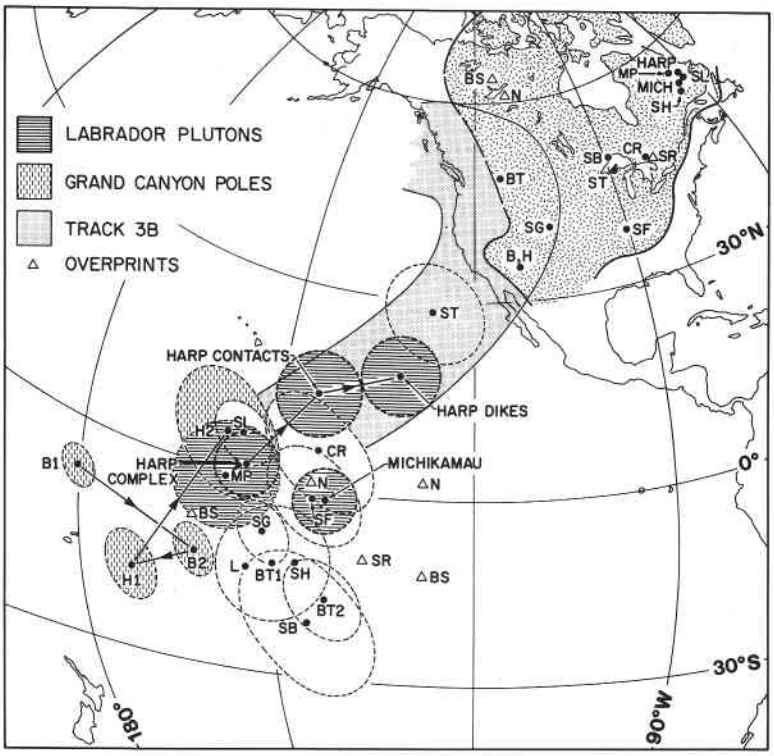


FIG. 11. Pole positions for interval 1200–1500 Ma. The sampling localities and corresponding poles are labelled as follows. *B1* and *B2* are two results from the Bass Limestone, and *H1* and *H2* two results from the Hakatai Shales of the Grand Canyon Supergroup in the stratigraphic order indicated (Elston and Gromme 1974). *BT1* and *BT2* Belt Supergroup (Collinson and Runcorn 1960; Evans *et al.* 1975). *CR* Croker Island Complex (Palmer 1969) Rb–Sr isochron 1475 Ma. *MP* Mistastin pluton (Fahrig and Jones 1976) Rb–Sr whole-rock age 1346 ± 30, see text where reasons for believing the age to be about 100 Ma older are given. *SB* Sibley Group (Robertson 1973) Rb–Sr isochron 1370 ± 72 Ma. *SF* St. Francois Mts. (Hsu *et al.* 1966) Rb–Sr isochrons 1400, U–Pb zircon ages 1500 (Bickford and Mose 1975). *SG* Sherman Granite (Eggler and Larson 1968) isochron 1410 Ma. *SL* Seal Lake Group red beds (Roy and Fahrig 1973) Rb–Sr isochron 1350 ± 92 Ma. *ST* South Trap Range Lavas, earliest Keweenawan and therefore older than −1150 Ma, magnetostratigraphy suggests age of about 1220 Ma (Irving and McGlynn 1976*a*). Added for sake of completeness are poles as yet inaccurately determined for the *L* Laramie Anorthosite one site only (Hoblitt and Larson 1975), and the *SH* Shabogamo Gabbro (Fahrig *et al.* 1974) sampled at 3 sites at a distance of over 80 km from the Grenville Front and outside the thermal effect of post-Grenville uplift (Ueno and Irving 1976). Also marked are overprints from *N* Nonacho Group (McGlynn *et al.* 1974), *BS* Big Spruce Complex (Irving and McGlynn 1976*b*), and *SR* Spanish River complex (Robertson and Watkinson 1974).

to each collecting site. The Harp dikes have steeper inclinations than the Harp Lake Complex, and the mean direction of the contact rocks is intermediate between them. All three are significantly different from one another. The difference between the dikes and the complex indicates that the magnetization of the complex predates the intrusion of the Harp dikes at about 1350 Ma. The presence of reversals and the high blocking temperatures and remanent coercive forces in the anorthositic rocks indicate that the magnetization was acquired at the time of initial cooling about 1450 Ma. At the time of initial cooling the complex was buried beneath cover at a depth of 10 km or more (R. F. Emslie, in preparation). The country rocks have broadly similar directions (Fig. 9), but they are not exactly the same, as is required for a successful contact test. We suggest therefore, that the country rocks became magnetized during uplift some considerable time after intrusion of the complex.

The poles available from Laurentia for the interval about 1500 to 1300 Ma are plotted in

Fig. 11. There are about 20 accurate determinations. The pole for the Harp Lake Complex agrees very well with that for the Mistastin pluton. The Mistastin pluton is dominantly adamellite, similar to the adamellites of the Harp Lake Complex. A whole-rock Rb–Sr isochron age of 1346 ± 15 Ma has been reported by Marchand and Crockett (1974) who point out that the moderately high $^{87}Sr/^{86}Sr$ initial ratio (0.7082) is indicative of crustal contamination. The 'isochron' may therefore be a mixing line. It is perhaps significant that the products of Elsonian magnetism in southeastern Missouri consistently give Rb–Sr isochron ages about 100 Ma younger than zircon ages from the same rocks (Bickford and Mose 1975). We have given our reasons above for believing that the Harp Lake anorthosite and adamellite pole has an age of about 1450 Ma; the pole from the Mistastin pluton is identical within experimental uncertainties and we suggest therefore that the Mistastin pluton and Harp Lake Complex were emplaced at about the same time. Although their radiometric ages are very similar the poles for the neighbouring Harp Lake Complex and Michikamau Intrusion are slightly different (Fig. 11). They have slightly different declinations (Fig. 1). This difference could be a real polar change occurring in the interval represented by the uncertainties in the radiometric age determinations, or it could be caused by subsequent relative rotation of the two bodies, for which, at present, there is no geological evidence either for or against. However there is very good *general* agreement between the poles for the Harp Lake Complex, the Mistastin pluton (*MP*), the Michikamau Intrusion, the Croker Island Complex (*CR*), the Sherman Granite (*SG*) and the rocks of the St. Francois Mountains (*SF*). The ages of these bodies very probably fall within the limits 1450–1500 Ma. They span a distance exceeding 3000 km. The poles for the Sibley Group and the Beltian rocks fall somewhat to

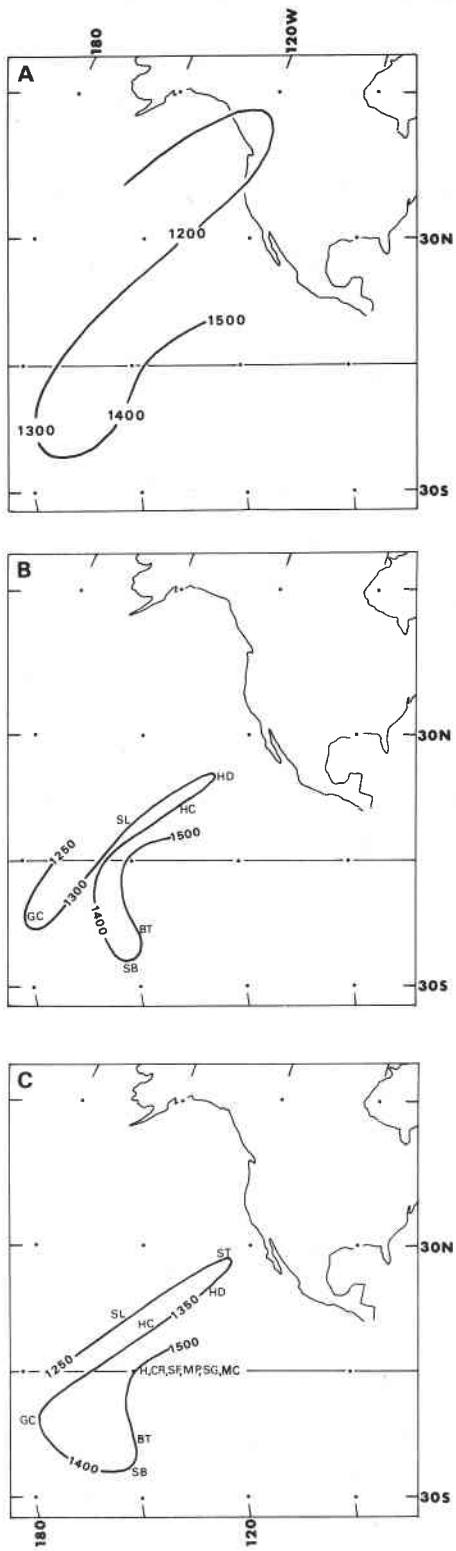


FIG. 12. Polar paths for interval 1200–1500 Ma. The simple loop suggested by Du Bois (1962), Spall (1971), Irving and McGlynn (1976*a*) and other workers is shown in *A*. More complex polar paths compatible with more recent information are shown in *B* and *C*. The poles are labelled as before, with the addition of the following: *GC* the lower part of the Grand Canyon Supergroup, *H* Harp Lake Complex, *MC* Michikamau Intrusion, *HD* Harp dikes, *HC* Harp Complex contact aureole and *BT* Belt Supergroup.

the south, and they may be a little younger, about 1400 Ma (see legend, Fig. 12). There may have been a polar shift from a position near to the equator at about 1500 Ma, to about 20° south at about 1400 Ma. The poles for the two lower formations (Bass Limestone and Hakatai Shales) of the Grand Canyon Supergroup fall to the west.

The pole positions for the time interval 1250–1500 Ma occupy a large area in the central Pacific and they show that Laurentia was near the equator during this time interval. In their attempts to draw pole paths most reviewers have followed Spall (1971) and drawn a simple polar loop, similar to that shown in Fig. 12 (*A*), but Elston and Gromme (1974) have obtained sequential poles from the Grand Canyon Supergroup which indicate that the variations may be more complex, and there are now several other results that are difficult to reconcile with the single loop concept.

An essential feature of the single loop concept is the existence of an ascending (to more northerly latitudes) polar track between about 1250 and 1150 Ma. This is the ascending limb of the Logan Loop (Robertson and Fahrig 1971), which is referred to here as track 3B, following the nomenclature of Irving and McGlynn (1976*a*). Initially, the evidence for track 3B was based mainly on the pole corresponding to the dominant magnetization of the Abitibi dikes (Larochelle 1966; Robertson and Fahrig 1971), but subsequently it has been shown that these dikes were intruded before 2000 Ma (Gates and Hurley 1973), and that the dominant magnetization was in fact acquired at the time the dikes were intruded (Irving and Naldrett 1977). Hence, the Abitibi pole is no longer relevant to the discussion of track 3B. Evidence that may be taken to support track 3B has been supplied (Elston and Gromme 1974) by a polar sequence observed from the Grand Canyon Supergroup (see Fig. 7 *in* Irving and McGlynn 1976*a*) but this sequence consists of older and younger groups of poles connected by an insecurely based pole from the Shinumo quartzite (9 samples only). The only remaining pole that actually falls in track 3B is that from the upper sequence of the South Trap Range lavas of the Keweenawan Peninsula (pole *ST* of Fig. 11, (Books 1972)). The age of this pole is uncertain but the rocks themselves are considered to be among the oldest of the Keweenawan units. Thus the evidence for the existence of track 3B has always been poorly documented (Irving and McGlynn 1976*a*, p. 445) and is now considered to be slender. Nevertheless, let us start with the assumption that track 3B exists, because it is essential to the single loop concept. The pole for the Harp dikes falls in track 3B but the estimated age for them (about 1350 Ma) is more than 100 Ma older than track 3B; the results from the Harp dikes do not conform well to the supposed age of track 3B. The second problem encountered when attempting to accommodate the results with a single polar loop concerns the pole for the Seal Lake red beds (*SL* of Fig. 11). It is indistinguishable from the pole for the Harp Lake Complex, although the Seal Lake Group is much younger than the complex, and is probably comparable in age to the Harp dikes. The third problem is concerned with the relative positions of the poles for the Seal Lake red beds and the Mackenzie igneous rocks. Their probable ages are 1350 and 1225 Ma respectively, whereas if they lie on the track 3B they ought to have the opposite time order. The single loop concept is therefore considered to be an inadequate explanation of the 1500 to 1250 Ma results.

In Fig. 12 (*B*,*C*) attempts are made to form the poles into an ordered sequence consistent, considering the uncertainties, with the geological relationships. The above discussion casts doubt on the existence of track 3B, so that the traditional approach used by Du Bois (1962) and all subsequent workers to connect the older (1500–1250 Ma) poles with the younger (1150–1000 Ma) Keweenawan poles may be in error. Consequently, we have made no assumptions about how this forward connection should be made. In Fig. 12(*B*) the path has a W-shape. The poles from the Grand Canyon Supergroup are connected by a simple loop that is intended to show their possible general age relationships only; the polar sequence of Fig. 11 is a stratigraphic sequence but it may not be a true magnetic age sequence because the magnetizations may not be original. In Fig. 12(*C*) a possible alternative S-shaped path is given, and this by no means exhausts the possibilities. Only one conclusion seems to be clear—that a simple polar loop does not adequately explain the information now available.

As already noted, there are about 20 well-determined poles for the interval 1500–1300 Ma from Laurentia. But it is as yet unknown whether

the poles of Fig. 11 are north or south poles because their forward connection with younger poles is not yet unambiguously determined. For *all* other Precambrian Shields of the world there are only 4 pole determinations. There are 3 poles from Australia: from the Morowa lavas, 1390 ± 140 Ma (Giddings 1975), from the Lunch Creek lopolith, 1498 ± 70 Ma (Duff and Embleton 1976), and from a set of metadiabase dikes from the Gawler Block (Giddings and Embleton 1976); and there is one pole from Africa, from the Pilansberg dikes, Rb–Sr age 1310 ± 60 Ma, Gough (1956). On the basis of such a scanty and unbalanced record, it is not possible to construct realistic polar wander curves for the other continents in this time interval. Even for the Laurentian Shield where the data base is several times greater than that for any other Precambrian Shield, the polarity of the poles is uncertain—an ambiguity of 180°! Therefore, the relative positions of the major Precambrian Shields during this interval cannot be unequivocally determined paleomagnetically.

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